

An Analysis of Precision and recall

Precision and recall

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}$$

Precision and recall -- Part 1

Thus the precision has an explicit dependence on r . Starting with balanced classes at $r = 1$ and gradually decreasing r , the corresponding precision will decrease, because the denominator increases. Another metric is the predicted positive condition rate (PPCR), which identifies the percentage of the total population that is flagged. For example, for a search engine that returns 30 results (retrieved documents) out of 1,000,000 documents, the PPCR is 0.003%.

Precision and recall -- Part 2

$$F_\beta = (1 + \beta^2) \cdot \text{precision} \cdot \text{recall} / (\beta^2 \cdot \text{precision} + \text{recall})$$

Precision and recall -- Part 3

Quantitative finance. A model may output a continuous buy/sell signal proportional to its conviction for each asset at a given moment. Applying a fixed threshold (e.g. $\text{signal} > 0.1 = \text{buy}$) discards conviction magnitude and conflates strong and mild signals, limiting the model's ability to correctly distinguish a strong buy from a mild one. DNA methylation analysis. Sequencing tools detect differentially methylated regions whose underlying signal is continuous, typically ranging from -1 to $+1$. Forcing this into a binary framework introduces similar distortions by treating all positive deviations as equivalent regardless of magnitude.

Precision and recall -- Part 4

Predicted positive condition rate = $\frac{TP + FP}{TP + FP + TN + FN}$. According to Saito and Rehmsmeier, precision-recall plots are more informative than ROC plots when evaluating binary classifiers on imbalanced data. In such scenarios, ROC plots may be visually deceptive with respect to conclusions about the reliability of classification performance. Different from the above approaches, if an imbalance scaling is applied directly by weighting the confusion matrix elements, the standard metrics definitions still apply even in the case of imbalanced datasets. The weighting procedure relates the confusion matrix elements to the support set of each

considered class.

Precision and recall -- Part 5

Note, sums may be replaced with integrals for continuous distributions depending on other continuous variables (i.e. $\sum \rightarrow \int$). These definitions reduce to the classical binary versions when $x_i, y_i \in \{0, 1\}$. However, continuous metrics reveal a deeper mathematical elegance. Consider a completely wrong ML model with random signal. Traditional binary metrics, blind to magnitude and sign, can still award such a model positive precision, recall and/or F-score (even 25% or higher for dense signal). The continuous formulas, by contrast, return exactly 0 for precision, recall, and F-score in this situation – because when expected and observed signals do not correlate, the numerator terms cancel to zero.

Precision and recall -- Part 6

Precision vs. recall Both precision and recall may be useful in cases where there is imbalanced data. However, it may be valuable to prioritize one metric over the other in cases where the outcome of a false positive or false negative is costly. For example, in medical diagnosis, a false positive test can lead to unnecessary treatment and expenses. In this situation, it is useful to value precision over recall. In other cases, the cost of a false negative is high, and recall may be a more valuable metric. For instance, the cost of a false negative in fraud detection is high, as failing to detect a fraudulent transaction can result in significant financial loss.

Precision and recall -- Part 7

Two other commonly used F_2 measure, which weights recall higher than precision, and the $F_{0.5}$ measure, which puts more emphasis on precision than recall. The F-measure was derived by van Rijsbergen (1979) so that F_β "measures the effectiveness of retrieval with respect to a user who attaches β times as much importance to recall as precision". It is based on van Rijsbergen's effectiveness measure $E_\alpha = 1 - \frac{1 - \alpha}{1 + \alpha} \frac{P}{R}$, the second term being the weighted harmonic mean of precision and recall with weights $(\alpha, 1 - \alpha)$. Their relationship is $F_\beta = 1 - E_\alpha$ where $\alpha = \frac{1}{1 + \beta^2}$.

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Precision and recall -- Part 8

Multi-class evaluation Precision and recall values can also be calculated for classification problems with more than two classes. To obtain the precision for a given class, we divide the number of true positives by the classifier bias towards this class (number of times that the classifier has predicted the class). To calculate the recall for a given class, we divide the number of true positives by the prevalence of this class (number of times that the class occurs in the data sample). The class-wise precision and recall values can then be combined into an overall multi-class evaluation score, e.g., using the macro F1 metric.

Precision and recall -- Part 9

Both precision and recall are therefore based on relevance. Consider a computer program for recognizing dogs (the relevant element) in a digital photograph. Upon processing a picture which contains ten cats and twelve dogs, the program identifies eight dogs. Of the eight elements identified as dogs, only five actually are dogs (true positives), while the other three are cats (false positives). Seven dogs were missed (false negatives), and seven cats were correctly excluded (true negatives). The program's precision is then $5/8$ (true positives / selected elements) while its recall is $5/12$ (true positives / relevant elements). Adopting a hypothesis-testing approach, where in this case, the null hypothesis is that a given item is irrelevant (not a dog), absence of type I and type II errors (perfect specificity and sensitivity) corresponds respectively to perfect precision (no false positives) and perfect recall (no false negatives). More generally, recall is simply the complement of the type II error rate (i.e., one minus the type II error rate). Precision is related to the type I error rate, but in a slightly more complicated way, as it also depends upon the prior distribution of seeing a relevant vs. an irrelevant item. The above cat and dog example contained $8 - 5 = 3$ type I errors (false positives) out of 10 total cats (true negatives), for a type I error rate of $3/10$, and $12 - 5 = 7$ type II errors (false negatives), for a type II error rate of $7/12$. Precision can be seen as a measure of quality, and recall as a measure of quantity. Higher precision means that an algorithm returns more relevant results than irrelevant ones, and high recall means that an algorithm returns most of the relevant results (whether or not irrelevant ones are also returned).